Module 54

Backup & Recovery/4: Recovery/3

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Module Recap

- Learnt how Hot backup of transaction log helps in recovering consistent database

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- Learnt how Hot backup of transaction log helps in recovering consistent database
- Studied the recovery algorithms for concurrent transactions

Module Objectives

- To understand Recovery with Early Lock Release

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- To understand Recovery with Early Lock Release
- To understand how to plan for backup and recovery

Module Outline

- Recovery with Early Lock Release

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- Planning Backup and Recovery

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- Recovery with Early Lock Release
- Planning Backup and Recovery

References:

- Enterprise Systems Backup and Recovery: A Corporate Insurance Policy by Preston De Guise
- `https://www.veritas.com/information-center/data-backup-and-recovery` (Accessed 19-Aug-2021)
- `https://www.sqlshack.com` (Accessed 19-Aug-2021)

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- Hence, the only way to undo the effect of insertion of (V_1, R_1) is to execute a corresponding delete operation

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  - subtract amount deposited, to undo deposit (allows early lock release on bank balance)

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- Physical redo logging does not conflict with early lock release

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  - In contrast, logging of old-value and new-value information is called **physical logging**, and the corresponding log records are called **physical log records**

Operation Logging (2): Example

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Log Output Example

```
<T1, O1, operation-begin>
<T1, X, 10, K5>                    \
                        Physical redo of steps in insert
<T1, Y, 45, RID7>                  /
<T1, O1, operation-end, (delete I9, K5, RID7)>
```

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- Redo of operation (after crash) still uses physical redo information

Transaction Rollback

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e) Stop the scan when the record <T_i `start`> is found

f) Add a <T_i `abort`> record to the log

Transaction Rollback with Logical Undo (cont.)

Note:

- Cases (c) and (d) above can occur only if the database crashes while a transaction is being rolled back
- Skipping of log records as in case (d) is important to prevent multiple rollback of the same operation

Transaction Rollback with Logical Undo (Diagram)

Beginning of log

<T_0 start>
<T_0, B, 2000, 2050>
<T_0, O_1, operation-begin>
<T_0, C, 700, 600>
<T_0, O_1, operation-end, (C, +100)>
<T_1 start>
<T_1, O_2, operation-begin>
<T_1, C, 600, 400>
<T_1, O_2, operation-end, (C, +200)>
<T_0 , C, 400, 500>
<T_0, O_1, operation-abort>
<T_0, B, 2000>
<T_0, abort>
<T_1, commit>

If T_0 aborts before operation O_1 ends, undo of update to C will be physical

T_0 has completed operation O_1 on C, releases lower-level lock; physical undo cannot be done anymore, logical undo will add 100 to C

T_1 can update C since T_0 has released lower-level lock on C

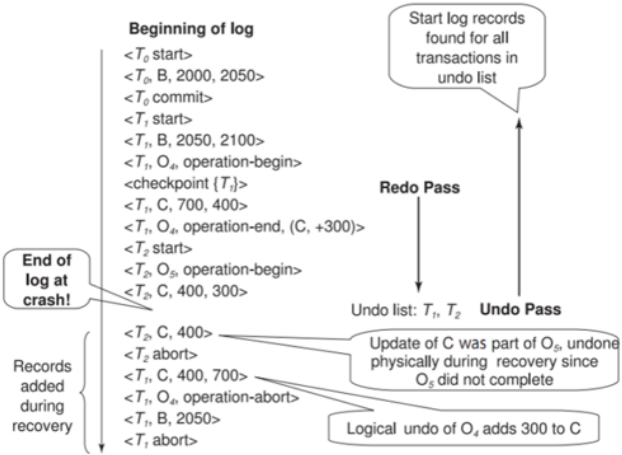
T_1 releases lower-level lock on C

T0 decides to abort

Logical undo of O_1 adds 100 to C

O_1 undo complete

Failure Recovery with Logical Undo (Diagram)



Transaction Rollback: Another Example

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```
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<T1, O1, operation-begin>
...
<T1, X, 10, K5>
<T1, Y, 45, RID7>
<T1, O1, operation-end, (delete I9, K5, RID7)>
<T1, O2, operation-begin>
<T1, Z, 45, 70>
    <- T1 Rollback begins here
<T1, Z, 45>                <- Redo-only log record during physical undo
                                (of incomplete O2)
<T1, Y, ..., ...>          <- Normal redo records for logical undo of O1
...
<T1, O1, operation-abort> <- What if crash occurred immediately after this?
<T1, abort>
```

Recovery Algorithm

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- **(Redo phase):** Scan log forward from last `<checkpoint L>` record till end of log

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- Now `undo-list` contains transactions that are **incomplete**, that is, have neither committed nor been fully rolled back

Recovery Algorithm with Logical Undo (2)

Recovery from system crash (cont.)

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Recovery Algorithm with Logical Undo (2)

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Recovery from system crash (cont.)

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- Stop scan when <T_i `start`> records have been found for all T_i in `undo-list`

Recovery Algorithm with Logical Undo (2)

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- When <T_i `start`> is found for a transaction T_i in `undo-list`, write a <T_i `abort`> log record.
- Stop scan when <T_i `start`> records have been found for all T_i in `undo-list`
- This undoes the effects of incomplete transactions (those with neither commit nor abort log records). Recovery is now complete.

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Deciding factors for having a Backup & Recovery setup.

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  - How often does your database get updated? For instance, if critical data is modified daily then you should make a daily backup schedule.
- **Speed:**
  - How much time do you need to back up or recover your files? Recovery speed is an important factor that determines the maximum possible time period that could be spent on database backup and recovery.

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Plan for Backup and Recovery (2)

Deciding factors for having a Backup & Recovery setup (cont.).

- **Equipment:**

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  - Do you have necessary equipment to make backups? To perform timely backups and recoveries, you need to have proper software and hardware resources.

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  - Do you have necessary equipment to make backups? To perform timely backups and recoveries, you need to have proper software and hardware resources.
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  - Who will be responsible for implementing your database backup and recovery plan? Ideally, one person should be appointed for controlling and supervising the plan, and several IT specialists (e.g. system administrators) should be responsible for performing the actual backup and recovery of data.

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- **Storing:**
  - Where do you plan to store database duplicates? In case of Online/Offsite storage you can recover your systems in case of a natural disaster. Storing backups on-site is essential to quick restore. But onsite storage has capacity bottlenecks and high maintenance costs.

Module Summary

- Recovery based on operation logging supplements log-based recovery

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